Results Report — All Methods Running

"What Crosses the Cut?" BP-Free Split-Learning Benchmark

Six methods running on GPU; MNIST, IID and non-IID

2026-08-30 · 4 rounds, seed 200, CUDA

1. Summary

All six methods now run end-to-end on GPU through the shared framework, on MNIST under both data settings (IID and α=0.5 non-IID), producing complete measurements for every method. This confirms the pipeline works: the same training loop, data pipeline, and metric logger run every method with no per-method special-casing, and the results separate the methods clearly. These are short 4-round runs to verify the pipeline, not final benchmark numbers.

2. Main results

Table 1 reports final test accuracy and total communication for each method, under IID (α=∞) and non-IID (α=0.5). **R** is the average operational rank across settings (lower is better).

Table 1: MNIST, 4 rounds. Accuracy (%) and communication (GB) per method and setting. Lower rank R is better.

	α=∞ **(IID)**		α=0.5		
Method	**Acc %**	**Comm GB**	**Acc %**	**Comm GB**	**R**
SplitFedv1	99.08	3.87	83.86	3.87	4.00
SplitFedv2	98.88	3.87	89.74	3.87	4.00
CSE-FSL	98.85	1.22	94.83	1.22	2.75
FSL-SAGE	98.83	0.67	**96.84**	0.67	**2.25**
HO-SFL	52.38	**0.16**	57.14	**0.16**	3.00
MU-SplitFed*	11.28	3.57	11.83	3.57	5.00

*MU-SplitFed was run from a third-party (HO-SFL repo) implementation, not the original authors' code — flagged pending its own validation.

What the numbers show:

- **The efficiency methods work as advertised.** FSL-SAGE reaches the same accuracy as the full-backprop baselines (∼98.8% IID) while using **0.67 GB vs. 3.87 GB** — about **5.8× less communication** — and ranks best overall (R=2.25). CSE-FSL is close behind (1.22 GB, R=2.75).
- **Non-IID separates the methods.** Under α=0.5 the baselines drop sharply (SplitFedv1 to 83.9%), while FSL-SAGE and CSE-FSL hold up best (96.8%, 94.8%) — the BP-free auxiliary-loss methods are more robust to heterogeneity here *and* cheaper.
- **HO-SFL trades accuracy for communication.** It uses the least communication by far (0.16 GB) but reaches only ∼52–57% accuracy in 4 rounds — a clear point on the trade-off, not a failure.
- **MU-SplitFed is flat at chance** (∼11%, = 1/10 on MNIST). Unlike HO-SFL, which climbs across rounds, MU-SplitFed's accuracy and loss are *flat* over all four rounds — the signature of an implementation/configuration problem, not slow convergence. It is the third-party (HO-SFL-repo) port, not the authors' code, so it is flagged for investigation rather than treated as a result.

3. How R is calculated

R aggregates each method's performance across settings into one comparable number (lower = better overall). It is computed by ranking, not by averaging raw values — so that accuracy (%) and communication (GB), which are on different scales, can be combined fairly. Four steps:

1. **Rank each metric within each setting.** In a given setting, rank the six methods on accuracy (higher = rank 1) and on communication (lower = rank 1) separately. Ties share the average rank.

2. **Average the metric ranks → per-setting rank.** For each method, average its accuracy-rank and communication-rank in that setting.
3. **Average across settings → R.** Average the per-setting ranks over IID and α=0.5.
4. **Lower R wins.**

Worked example — FSL-SAGE (the best, R=2.25):

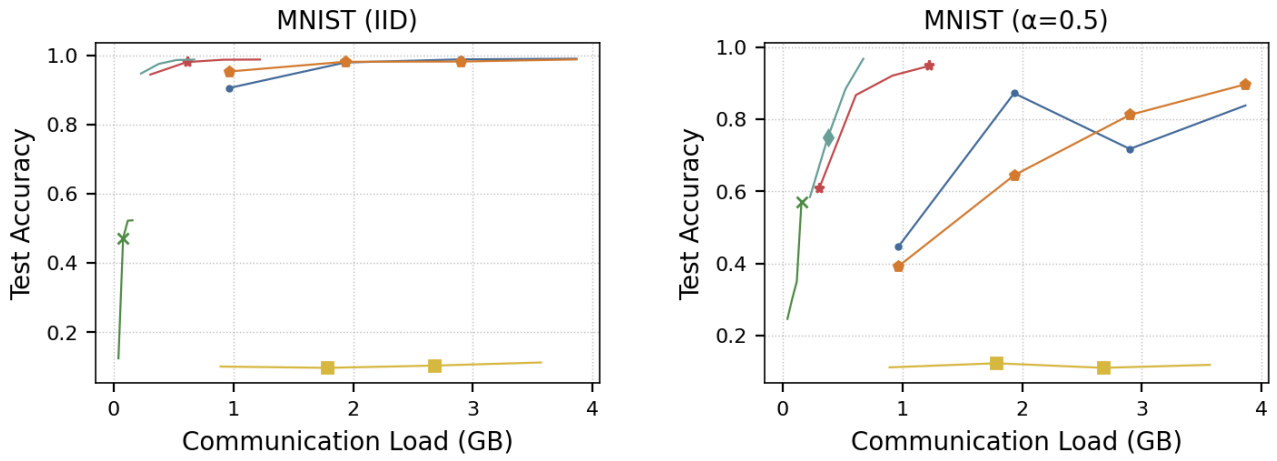
Setting	Acc rank	Comm rank	Per-setting avg
IID (α=∞)	4 (98.83%)	2 (0.67 GB)	(4+2)/2 = 3.00
α=0.5	1 (96.84%)	2 (0.67 GB)	(1+2)/2 = 1.50
	R = (3.00 + 1.50)/2 =		**2.25**

In IID, FSL-SAGE is only 4th on accuracy (the baselines edge it out) but 2nd on communication; under α=0.5 it is 1st on accuracy and still 2nd on communication. Averaging those gives R=2.25 — the lowest, because it is consistently strong on *both* axes across *both* settings. By the same procedure: CSE-FSL 2.75, HO-SFL 3.00, SplitFedv1 and SplitFedv2 4.00, MU-SplitFed 5.00.

Note: R here is computed over the two metrics available in these runs (accuracy, communication). In the full benchmark the same procedure extends to all metrics (adding memory and latency). R is provisional while MU-SplitFed is unresolved — its worst-place ranks push every other method's R down slightly, so the ranking will shift once it is fixed or dropped.

4. The trade-off, visualized

The Pareto view (Figure 1) plots accuracy against communication: up and to the left is better. It makes the ranking visible — FSL-SAGE and CSE-FSL sit in the strong region (high accuracy, low communication), the full-backprop baselines pay much more communication for similar or lower accuracy, and HO-SFL / MU-SplitFed occupy the low-cost / low-accuracy corner.



(a) IID (α=∞)

(b) Non-IID (α=0.5)

Figure 1: Test accuracy vs. communication load, MNIST. Up-and-left is better. FSL-SAGE and CSE-FSL dominate the efficient region; the baselines cost far more communication for comparable accuracy.

5. What this confirms

1. **All six methods run on GPU** through one shared harness, on two data settings, with all metrics logged — the full pipeline works, not just a single method.
2. **The benchmark discriminates.** The methods span ∼99% to ∼11% accuracy and 0.16 to 3.87 GB communication — clearly separating efficient methods, trade-offs, and a non-converging case.

3. **The communication metric captures the core story:** the efficient BP-free methods (FSL-SAGE, CSE-FSL) using a fraction of the baselines' communication — exactly what the benchmark exists to measure.

6. Notes and next steps

- **MU-SplitFed (∼11%)** is flat at chance across all four rounds (accuracy and loss do not move) — this points to the third-party port or a config issue, not to needing more training. Fix: run it standalone for many more rounds to confirm, and if it stays flat, obtain the authors' code or drop it. Flagged, not dropped.
- **HO-SFL** genuinely learns but slowly (52–57% in 4 rounds, curve still rising); worth confirming with more rounds.
- These are **4-round pipeline-verification runs on MNIST**, on our fixed GPU setup (identical for all methods). Communication is hardware-independent; latency is a relative within-setup figure.
- **Datasets to add:** beyond CIFAR-10 / CIFAR-100, the benchmark will include **STL-10** (higher-resolution natural images) and **MedMNIST** (medical domain-shift), broadening the comparison beyond a single data domain.
- **Next:** scale to CIFAR-10 / CIFAR-100 (then STL-10 and MedMNIST) with longer training to produce the actual benchmark table.

Bottom line: the code works. All six methods run on GPU across IID and non-IID MNIST, all metrics log, and the results already show the expected pattern — FSL-SAGE and CSE-FSL match baseline accuracy at a fraction of the communication. Ready to scale to CIFAR.